

Shriners Gait Model Setup Using Pelfix Option (Currently Vicon Use Only)

1. Moving VST files to correct location for Vicon (Vicon Skeletal Models)

- a. Copy the .vst files from the vsts folder in the GitHub project (ViconClinicalGait/vsts/)
 - i. Current vsts included with the project are:
 1. SMACNet no KAD Dypstick.vst (Standard Model)
 2. SMACNet Foot Model Dypstick no KAD (Multi-segment Foot Model)
 - ii. Note: The markersets used for these vary slightly from the original models. You can review the ReadMe file to see the highlighted differences.
- b. Paste into your Vicon ModelTemplates folder
 - i. Example filepath
C:\Users\Public\Documents\Vicon\Nexus2.x\ModelTemplates
- c. Note: All our vsts for the Pelfix option include the term “Dypstick” to distinguish it from the other models.
 - i. Dypstick (often referred to as a pointer) is the device we use to apply pelfix options to create virtual markers (commonly used for ASIS markers for patients with significant soft tissue envelope around the hips)
- d. For more in-depth information about the Dypstick, see the additional user guide provided (5.Dypstick_Pointer_Pelfix)
 - i. The rest of this guide will assume that you have a working pointer that functions in the same manner as this dypstick

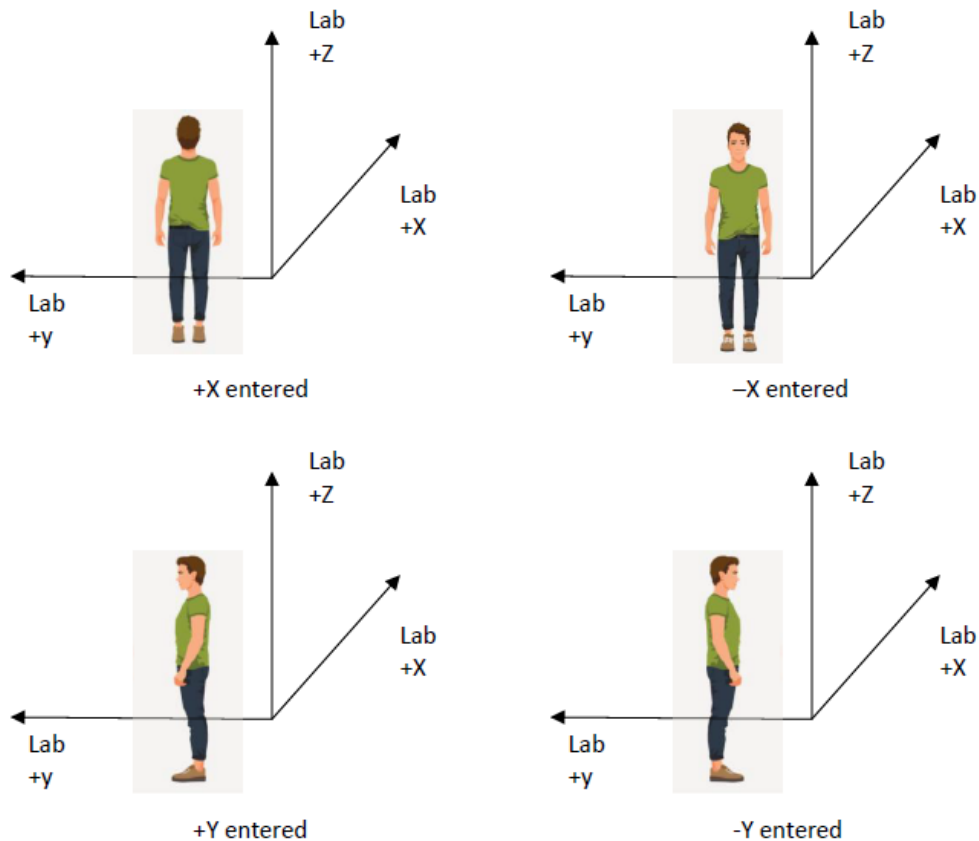
2. Moving Pipeline files to correct location for Vicon

- a. Copy the .Pipeline files from the Utils folder in the GitHub project (ViconClinicalGait/Utils/Pipeline)
 - i. Current Pipelines included with the project are:
 1. Py3_Static
 2. Py3_Dynamic
 3. Py3_SittingFoot_Static_BF
 4. Py3_Static_AFO
- b. Paste into your Vicon Pipelines folder
 - i. Example filepath
C:\Users\Public\Documents\Vicon\Nexus2.x\Configurations\Pipelines

3. Moving Python Code to the correct location for Vicon

- a. Copy the .py files from the Py3_ShrineGaitModel folder in the GitHub project (ViconClinicalGait/Py3_ShrineGaitModel)
 - i. Current Python files included with the project are:
 1. Py3_CreateGCD
 2. Py3_DynamicMain
 3. Py3_GaitModules

4. Py3_MathModules
 5. Py3_StaticMain
 6. Py3_SittingFootStaticMain.py
 7. Py3_WriteHeelToeMarkers.py
 8. Py3_UserPreferences
 - b. Paste into a Vicon Python folder
 - i. Example filepath: C:\Users\Public\Documents\Vicon\Python
 - c. Possible errors in where these files are placed could impact their use
 - d. File location matters significantly, the Py3_UserPreferences.py file must be in the same directory(folder) as the Py3_StaticMain.py file for it to identify the appropriate file.
4. **Updating the User Preferences File:** Several items need to be reviewed in "UserPreferences.py" for each individual lab, these are the contained in the first 6 lines of code
 - a. The marker diameter needs to be entered for your lab
 - i. Most likely the diameter will be 9.5 mm
 - ii. If you don't use a base that adds thickness the values entered will be:
 1. self.MarkerDiameter = 9.5
 - iii. If you use a base, add the thickness of the base and multiply by 2. For example if you use a 2 mm base add the value of 4 as follows:
 1. self.MarkerDiameter = 9.5+4
 - b. For the next line of code (self.StaticForwardDirection =) you will need to know the global axis coordinate system for your lab.
 - i. Determine the global axis coordinate system
 - ii. Determine the direction that participants stand during the static trial (+X, -X, +Y, -Y). Enter the direction the participant stands with respect to the lab coordinate system:
 - iii. Here are 4 examples:



- iv. Enter the correct '+X' or '-X' or '+Y' or '-Y' for your lab and data collection protocol: e.g. `self.StaticForwardDirection = '-X'`
- c. The 3rd line of code is related to the selection of the hip model implementation (`self.HipModelName`)
 - i. The default for clinical gait analysis for Shriners Children's is 'HarringtonHybrid'
 - ii. 'HarringtonHybrid' uses regression equations to determine the x,y,z coordinates of the hip joint centers. The full set of equations use Pelvic Width (measured distance between ASIS during physical exam), Pelvic Depth (distance between Midpoint of ASIS markers and Midpoint of PSIS markers), and Leg Length from physical exam. **BOTTOM LINE:** leg length and ASIS-width need to be measured on physical exam and used.
- d. The 4th and 5th lines establish the rotation sequence for the pelvis and trunk. The 2 options are the orders ROT (Rotation-Obliquity-Tilt) or TOR (Tilt-Obliquity-Rotation). **'ROT' is the setting that should be used.**
 - i. `self.TrunkRotationSequence = 'ROT'`
 - ii. `self.PelvisRotationSequence = 'ROT'`
- e. The 6th line sets the shank coordinate system as Distal or Proximal. **The setting of 'Distal' should be used.** NOTE: Distal or Proximal must start with capital letter

- i. self.ShankCoordinateSystem = Distal
- f. Lines 7 to 71 establish the marker labels used in the VST files. The labels in the py3_UserPreferences.py file match these labels as needed, so this part of the file does not need to be updated unless you decide to change the marker labels.
- g. **Lines 73-75 are specific to the Dypstick/Pointer**
 - i. **Lines 73 and 74 define the names of the tip and tail markers**
 - ii. **Line 75 defines the distance between the head of the pointer and the first marker. Based on the device specifications we provided, our distance is 120mm. For those that want to use a similar device, adjust this distance as needed.**

5. Static Trial

- a. In session folder, create subject and attach the appropriate vst (Standard Model or Foot Model with Dypstick)
 - i. Under Subject Properties, enter all required anthropometrics (outlined in red)
- b. **Record a Static trial with the Dypstick/Pointer**
 - i. **Note:** It can be helpful to identify where the ASIS is with a sticker or other mark before starting the trial
 - ii. Start the static trial with the Dypstick/Pointer away from the patient
 - iii. After you start collecting, push the tip of the device into the first ASIS
 - 1. Make sure you are really pushing through the tissue to get as close as possible
 - iv. Push the back end of the device so the two markers come closer together (this action is used to mark the frame to use for initial placement of the virtual marker, so make sure the tip is still in the proper position at this point)
 - v. Repeat this process for the other ASIS position
 - vi. Take the device and step away from the patient and end the trial
 - vii. **Note:** The device only consists of two markers, so it is helpful to move it slowly during the collection, make sure the markers are occluded minimally by the user, and move within a single plane to prevent switching throughout the trial
 - 1. This is difficult to describe, but you will know what we mean as you try using the device.
- c. Open static
 - i. Run "Reconstruct" pipeline

The screenshot shows a software interface with two sections: 'Left' and 'Right'. Each section contains a list of anthropometric measurements with corresponding input fields. Red boxes highlight the following fields in both sections:

- General:** Bodymass (kg), Height (mm), InterAsisDistance (mm)
- Left:** LegLength (mm), AsisTrocanterDistance (mm), KneeWidth (mm), AnkleWidth (mm), VarValAngle (deg)
- Right:** LegLength (mm), AsisTrocanterDistance (mm), KneeWidth (mm), AnkleWidth (mm), VarValAngle (deg)

Other fields visible include CalcanealPitch (deg), HindfootProgression_relBiMal (deg), 1stRayPitch (deg), and ForefootProgression_relBiMal (deg).

1. It is harder to reconstruct and label when using the pointer, since all markers are not in an expected position
- ii. Check marker labels, and pay special attention to the pointer tip and tail markers to make sure they are not switching
 1. Sometimes it is easier to take another static instead of dealing with the switching in post-processing
- d. Run "Py_3 Static" pipeline
 - i. On the following screen:
 1. Fill out required patient/subject information (outlined in red)
 2. Match Static Frame Number to the selected frame number of the static trial (outlined in blue)
 3. Check that all required anthropometric measures are filled in (outlined in green)
 4. If you are using the foot model
 - a. Under "Foot Model" (outlined in purple), check the boxes for both feet
 - b. Make sure Hindfoot Varus/Valgus is populated
 - c. Other measures are optional (we highly recommend reviewing the 4.FootModelSOP if using the foot model)
 5. Under "Special Test Conditions", select appropriate boxes to indicate whether subject is plantigrade and whether subject is shod. If shod, enter sole delta in mm (change in height of shoe sole between toe box and heel (positive number or 0))
 - a. Note: If using the foot model, the subject MUST be plantigrade during static
 - i. If they cannot stand plantigrade, please see 4.FootModelSOP for information on how to perform a sitting static where the feet are plantigrade
 6. We recommend not changing the knee option setting. This model will work with a KAD (knee alignment device), but we have shifted to using medial and lateral knee markers
 7. Next to "Apply Pelvic Fix Option" at the bottom of the screen, select the "Triad" option.
 - a. There is an additional pelfix method that uses iliac markers, but the method included with this model uses the sacral triad
 - b. Make sure you are using the Dypstick version of the vst if you are planning to take advantage of this option

Static

Patient Information

First Name Last Name

Patient Number

Date Of Birth 01 Jan 1981

Trial Activity Static

Trial Modifier Barefoot

Static Forward Direction +X

Assistive Device None

Anthropometric Parameters

Static Frame Number 85

Static File Anthropometric File Static_BF_2025Consistency

Date Of Data Capture 17 Dec 2025

kg 80.0 Body Mass

mm 1800 Height

mm 248 ASIS-to-ASIS Distance

mm 995 995 Leg Length

mm 127 126 Knee Width

mm 78 78 Ankle Width

mm 80 80 ASIS-to-GT Distance

Foot Model

☒ Left ☒ Right

deg [-Val/+Var] 0 Hindfoot Varus/Valgus

deg [+Up/-Down] Calcaneal Pitch

deg [+Int/-Ext] Hindfoot Progression [rel. to Bimalleolar axis]

deg [-Up/+Down] First Metatarsal Pitch

deg [+Int/-Ext] Forefoot Progression [rel. to Bimalleolar axis]

Special Test Conditions

☒ Left ☒ Right Subject Plantigrade during Static Trial

☐ Subject Shod (with or without orthoses)

mm Left Right Difference in sole thickness between heel and toes (mm)

☐ N/A ☐ Iliac ☐ Triad Apply Pelvic Fix Option

Knee Option ☐ KAD ☒ M/L Markers

Save & Proceed

Quit

8. Click “Save & Proceed”

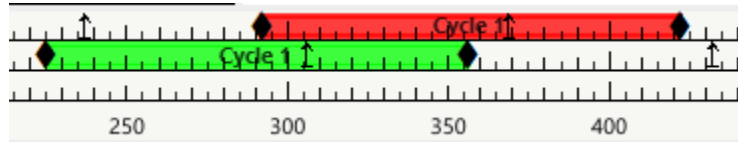
- Once you have gone past this screen, you should see the virtual markers have been placed in Nexus
 - We recommend reviewing the placement visually to see if it matches the positions, you expect
- Compare marker measurements to clinical measurements (The Shriners Gait Model uses clinical measurements to calculate joint centers and virtual markers). Click “Proceed”
 - Check Standing Posture measurements. If you are using the Standard Model, the Foot Segment measurements will be empty. Click “Save Results” and “Save PDF”. Click “Quit”
 - “Save Results” saves a static.py file that contains all of the static trial information needed to process the dynamic trials (marker relationships between technical and anatomical coordinate systems, anthropometrics, etc.)

2. "Save PDF" saves a pdf output report of the static trial you just processed, showing the marker measurement comparison, standing posture measurements, etc.

6. Dynamic Trials

- a. Medial knee and medial malleoli markers can be removed for dynamic trials
- b. Record dynamic walking trials in your capture volume

7. Processing Dynamic Trials

- a. Open dynamic walking trial
 - b. Run "Reconstruct & Label" pipeline
 - i. Note that reconstruct and label does not always work as well for this model since the ASIS markers are missing and added back later
 1. You may end up having to label manually often
 - ii. Check marker labels and gap fill as needed
 - iii. Crop trial as needed. The Shriners Gait Model requires all markers to be present on the first frame of the trial (this does not include the virtual ASIS markers that will be added in later)
 - c. Add in gait events
 - i. Label at least one full gait cycle as shown below
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- d. This model does support kinetics requiring force plates to be integrated with Vicon
 - i. Assign each force plate based on which foot strikes the plate (right or left)
 - e. Run "Py3_Dynamic" pipeline
 - f. You will see the ASIS markers added in while this pipeline is running.

8. Outputs

- a. The Py3_Dynamic pipeline will complete the calculations and save the results (kinematics, kinetics, etc.) as model outputs
- b. These outputs can be found in Vicon by:
 - i. clicking on the Subjects Tab
 - ii. clicking the drop down menu to the left of the subject name
 - iii. clicking the drop down menu to the left of Model Outputs
 - iv. and clicking the drop down menu to the left of the model output you are interested in
- c. The pipeline will also output a .GCD file
 - i. This is a type of text file that contains each model output time normalized to be represented as percentage of the gait cycle (101 points 0-100%)
 - ii. This text file can be used as a coding input to visualize the data
 - iii. No visualization code has been provided